

Cyclistic Bike-Share Analysis

How Casual Riders and Annual Members Use Cyclistic Bikes Differently

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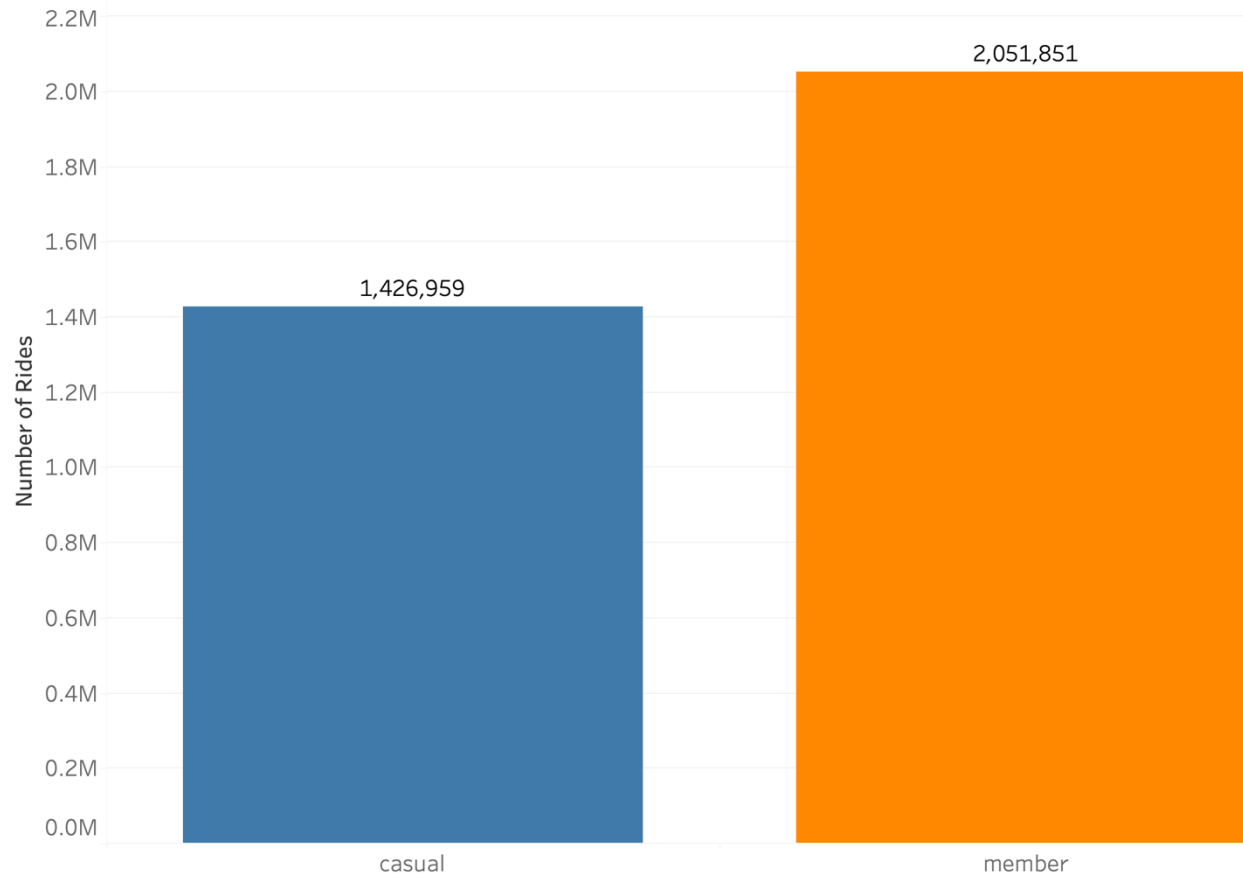
Business Task

- Analyze how casual riders and annual members use Cyclistic bikes differently and identify insights that can help convert casual riders into annual members.

Data and Methodology

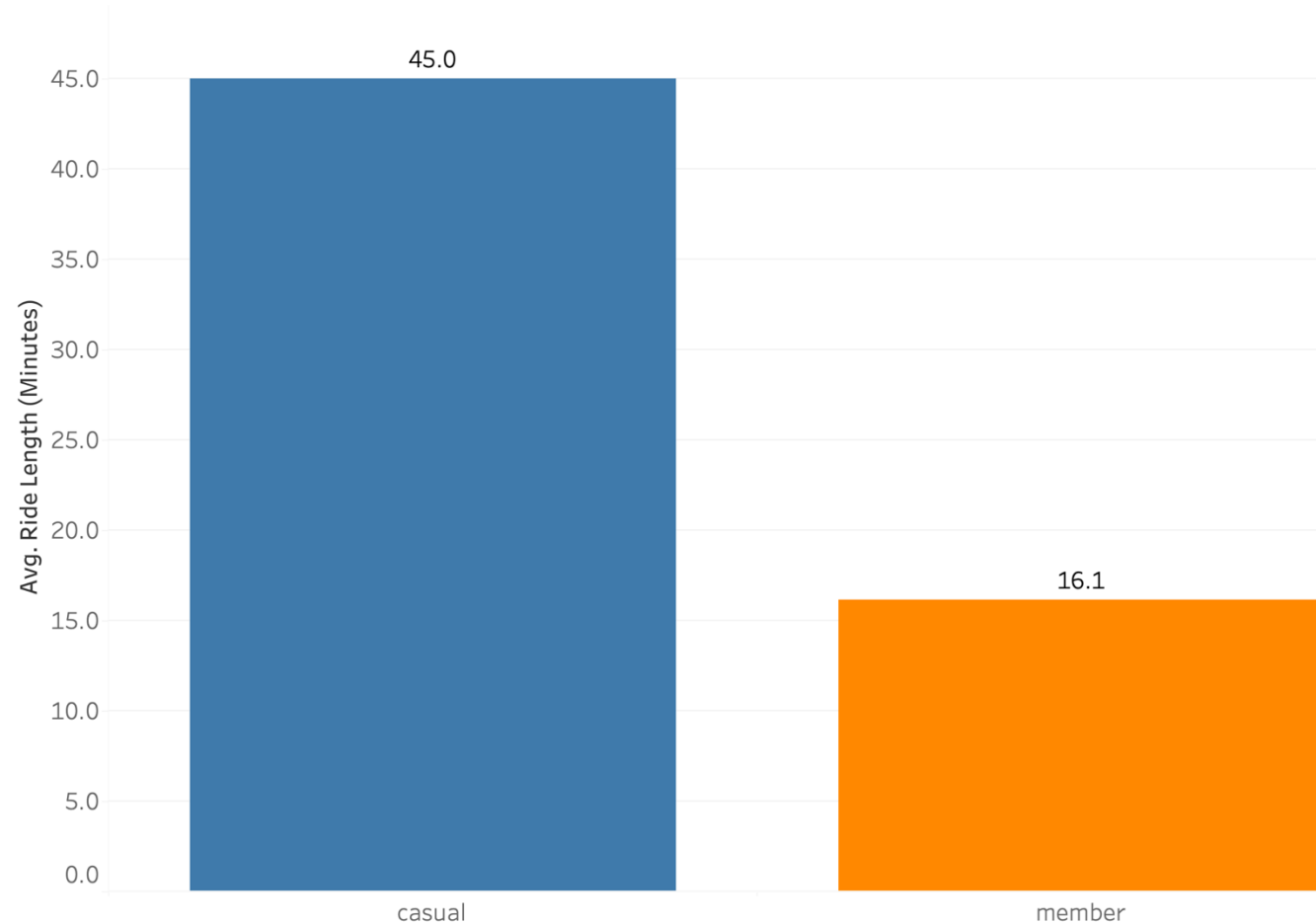
- **Data:** 12 months of Divvy bike-share trip data (April 2020–March 2021).
- **Preparation:** Combined 12 months CSV files using Excel Power Query.
- **Cleaning:** Checked data types and consistency, created ride-length/date-time fields, and removed rides with invalid durations (≤ 0).
- **Analysis:** Analyzed the cleaned dataset of **3,478,810 rides** using Excel summaries and PivotTables, then used Tableau for visualization.

Total Rides by Rider Type



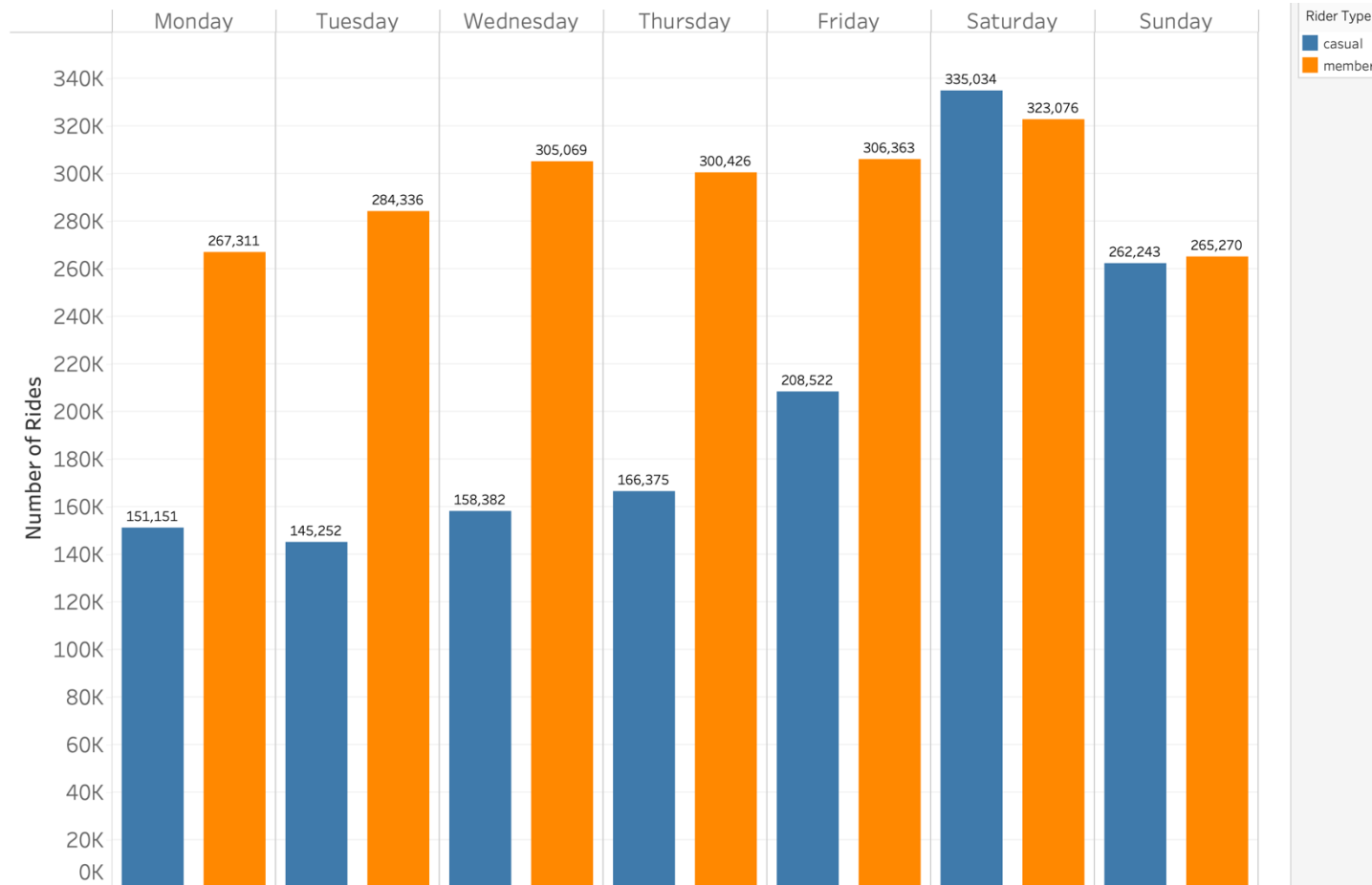
- Members completed **2,051,851 rides**, compared with **1,426,959 casual rides**.
- about **44% more rides** by members.

Average Ride Length by Rider Type



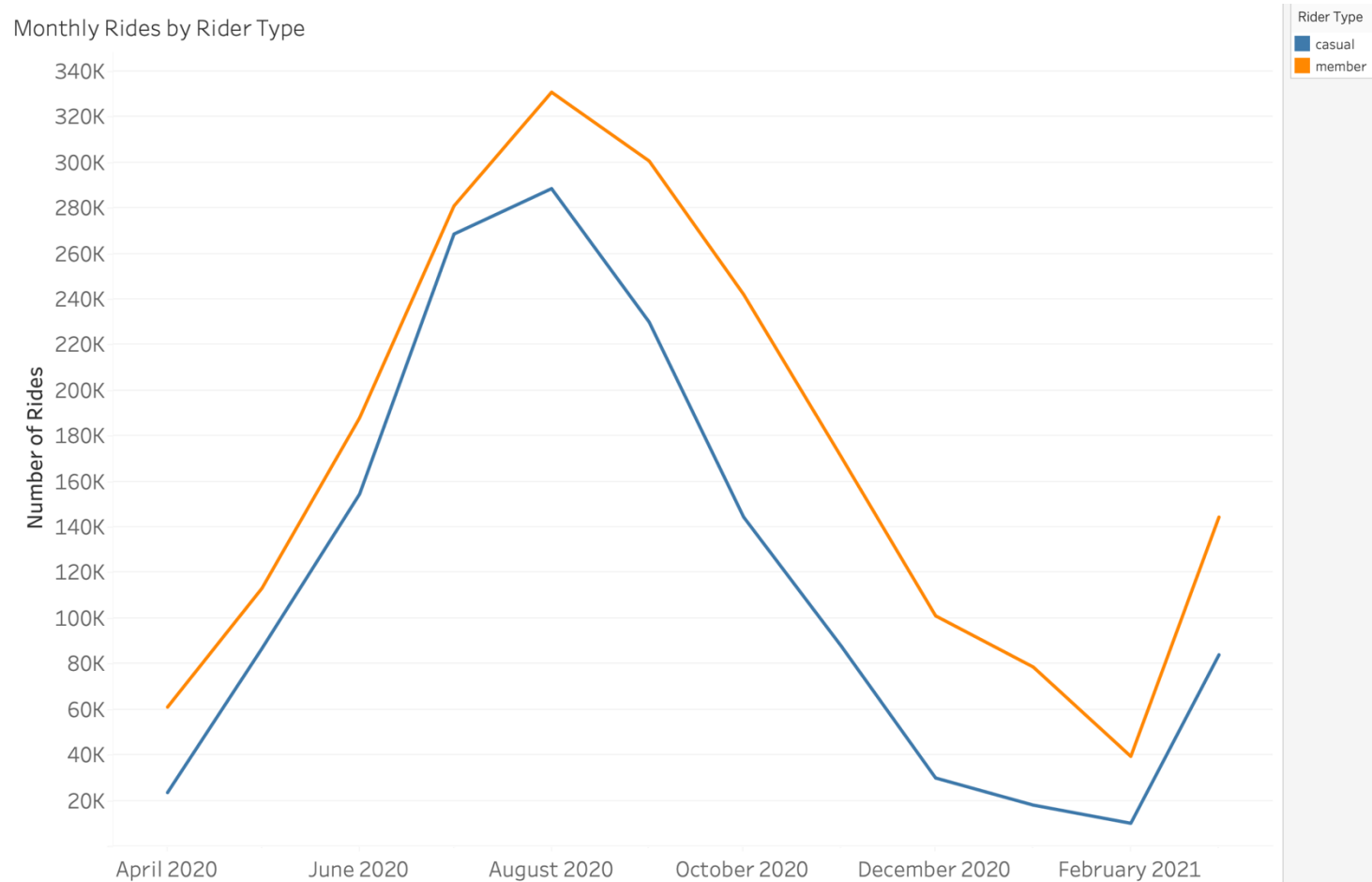
- Casual riders averaged 45.0 minutes per ride, compared with 16.1 minutes for members.
- Casual rides were nearly 3× longer on average.

Rides by Day of Week and Rider Type



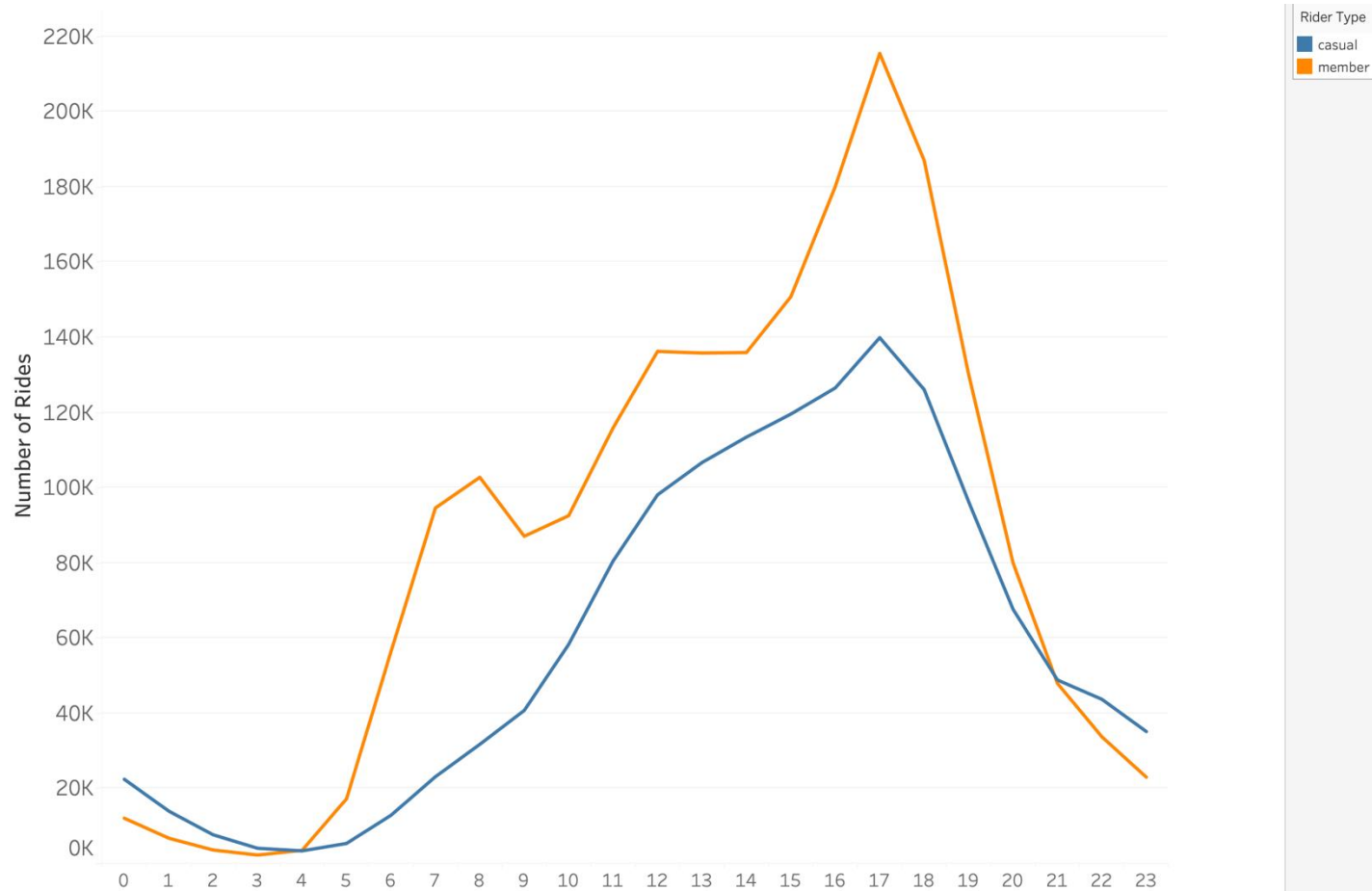
- Members take substantially more rides than casual riders from **Monday through Friday**.
- Saturday is the only day casual riders exceed members, with **335,034** casual rides vs. **323,076** member rides.
- Casual rides are strongly concentrated on the weekend.

Monthly Rides by Rider Type



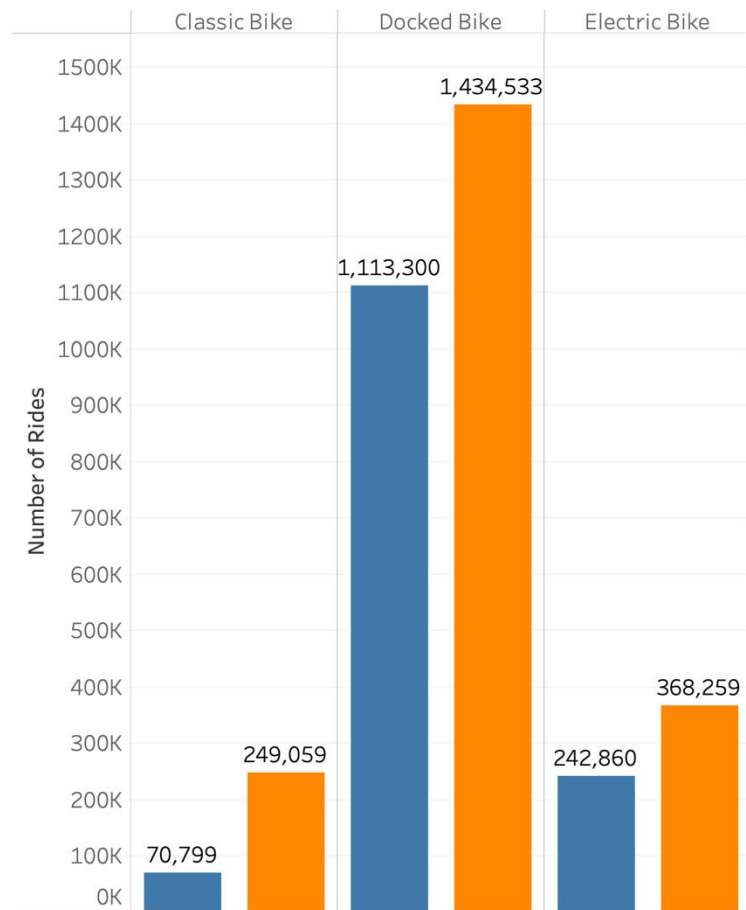
- Both rider types show strong **seasonal patterns**, with rides increasing through spring and summer and declining during winter.
- Ridership for both groups peaked in August 2020 — about **289K casual rides** and **331K member rides**.
- Members recorded more rides than casual riders in every month, while both groups reached their lowest levels in **February 2021**.

Rides by Hour and Rider Type



- Member rides show a pronounced late-afternoon peak, reaching their highest level around **5 PM (17)**.
- Casual rides increase more gradually throughout the day, **peaking around 5 PM**, but at a much lower level than member rides.
- Member rides show a noticeable **morning peak around 7–8 AM**, while casual ridership remains much lower during those hours.

Rides by Bike Type and Rider Type



- Docked bikes dominate ridership for both groups, accounting for about **1.11M casual rides** and **1.43M member rides**.
- Members recorded more rides across every bike type, with the largest numerical difference occurring for docked bikes.
- Electric bikes are the second-most-used bike type for both groups, while classic bikes account for the fewest rides.

Key Findings

- **Members ride more frequently overall**, completing about **44% more rides** than casual riders.
- **Casual riders take much longer trips**, averaging **45.0 minutes** compared with **16.1 minutes** for members.
- **Casual riding is more leisure-oriented**, with much stronger weekend activity, while member riding is stronger during weekdays and commuting hours.
- **Both groups are highly seasonal**, with ridership strongest in the warmer months and substantially lower during winter.

Recommendations

- **Target casual riders during weekends and warmer months.** Run membership campaigns when casual ridership is highest, especially on Saturdays and during spring and summer.
- **Promote membership around the value of frequent riding.** Use targeted offers to show casual riders how an annual membership could benefit them if they begin using Cyclistic more regularly.
- **Encourage casual riders to use Cyclistic beyond recreation.** Promote membership for weekday and commuting use, emphasizing Cyclistic as an option for routine transportation rather than only leisure trips.